Credit Card Approval Prediction

Project Description

Description

Credit Card Approval Prediction is an Artificial Intelligence and Machine Learning-based web application designed to predict whether a credit card application should be approved or rejected. The system helps banks and financial institutions make faster, smarter, and more accurate credit card approval decisions by analyzing applicant information using trained Machine Learning models

The application collects details such as Income Type, Employment Status, Annual Income, Family Status, Education Level, Housing Type, Occupation, Credit History, Existing Loan Status, Number of Dependents, and other financial and demographic attributes. These details are processed using Logistic Regression, Decision Tree, Random Forest, and XGBoost classification models. The best-performing model predicts whether the applicant is eligible for a credit card.

The system is developed using Flask, HTML, CSS, Bootstrap, JavaScript, Python, Scikit-learn, XGBoost, and IBM Watson Machine Learning. It provides a simple and user-friendly web interface that enables users to enter applicant details and receive instant credit card approval predictions.

The Credit Card Approval Prediction system minimizes manual effort, reduces application processing time, improves decision-making accuracy, and assists banks and financial institutions in evaluating applicants efficiently while reducing the risk of approving high-risk applications.

Scenarios

Scenario 1 – Loan Approval

A bank employee enters the details of an applicant with a stable income, good employment history, and a positive credit record. The system analyzes the applicant's information and predicts that the credit card application is approved within seconds.

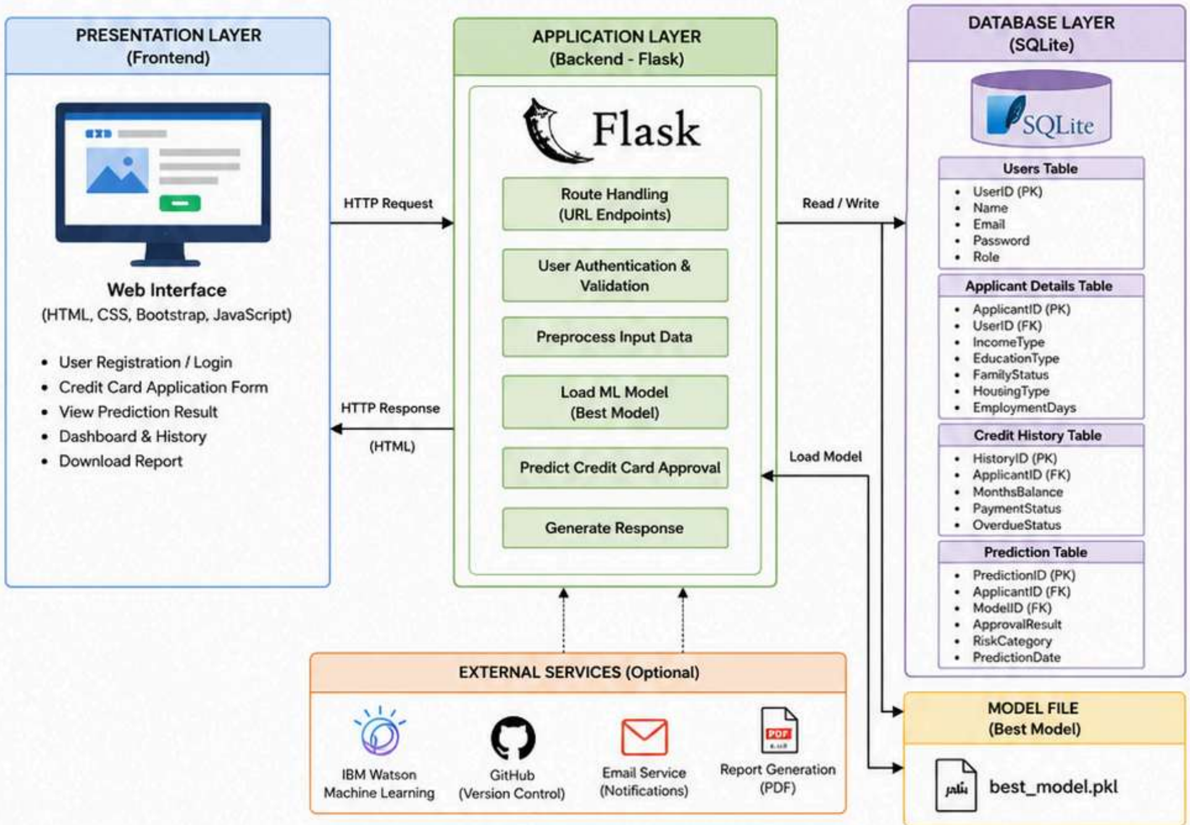
Scenario 2 – Credit Card Rejection

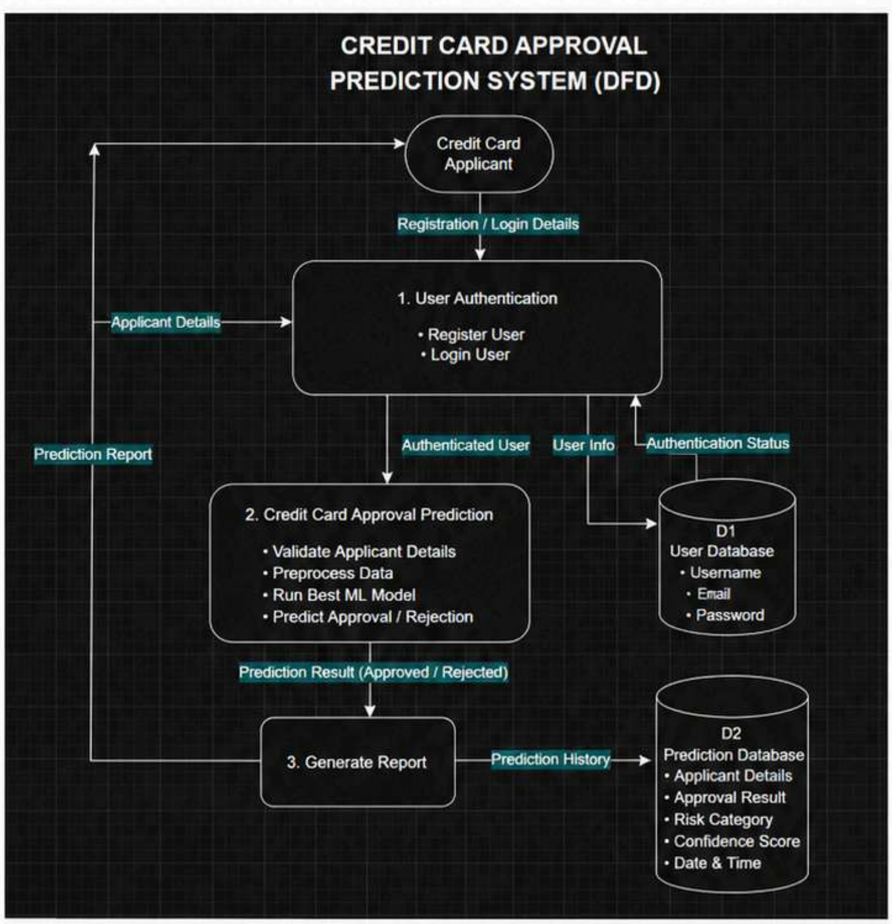
An applicant with low income, poor credit history, and multiple existing loan defaults submits a credit card application. The machine learning model predicts that the application should be rejected, helping the bank reduce financial risk and prevent potential defaults.

Scenario 3 – Faster Credit Card Application Processing

A bank receives thousands of credit card applications every day. The Credit Card Approval Prediction system automatically evaluates each application using machine learning, reducing manual effort, improving processing speed, and helping financial institutions make faster and more accurate approval decisions.

Technical Architecture:





Description

Credit Card Approval Prediction follows a three-tier architecture consisting of the Presentation Layer, Application Layer, and Database Layer.

The Presentation Layer provides a responsive user interface developed using HTML, CSS, Bootstrap, and JavaScript. Users can register, log in, enter applicant details, and submit credit card application information through an interactive web interface.

The Application Layer is developed using the Flask framework. It handles user authentication, validates applicant data, performs data preprocessing and feature engineering, loads the best-performing machine learning model selected from Logistic Regression, Decision Tree, Random Forest, and XGBoost, predicts whether the credit card application should be approved or rejected, and generates the final prediction.

The Database Layer uses SQLite to securely store user credentials, applicant details, credit history, machine learning model information, and credit card approval prediction records. All prediction results are saved for future reference and analysis.

This architecture ensures scalability, security, faster response time, efficient credit card application processing, and accurate decision-making for financial institutions.

Pre-requisites

- Python 3.10 or above
- Flask Framework
- Machine Learning using Scikit-Learn
- XGBoost
- Pandas
- NumPy
- Pickle
- Seaborn
- HTML
- CSS
- Bootstrap
- JavaScript
- SQLite Database
- VS Code
- Git & GitHub
- Jupyter Notebook
- Render Deployment
- IBM Watson Machine Learning (for cloud deployment)

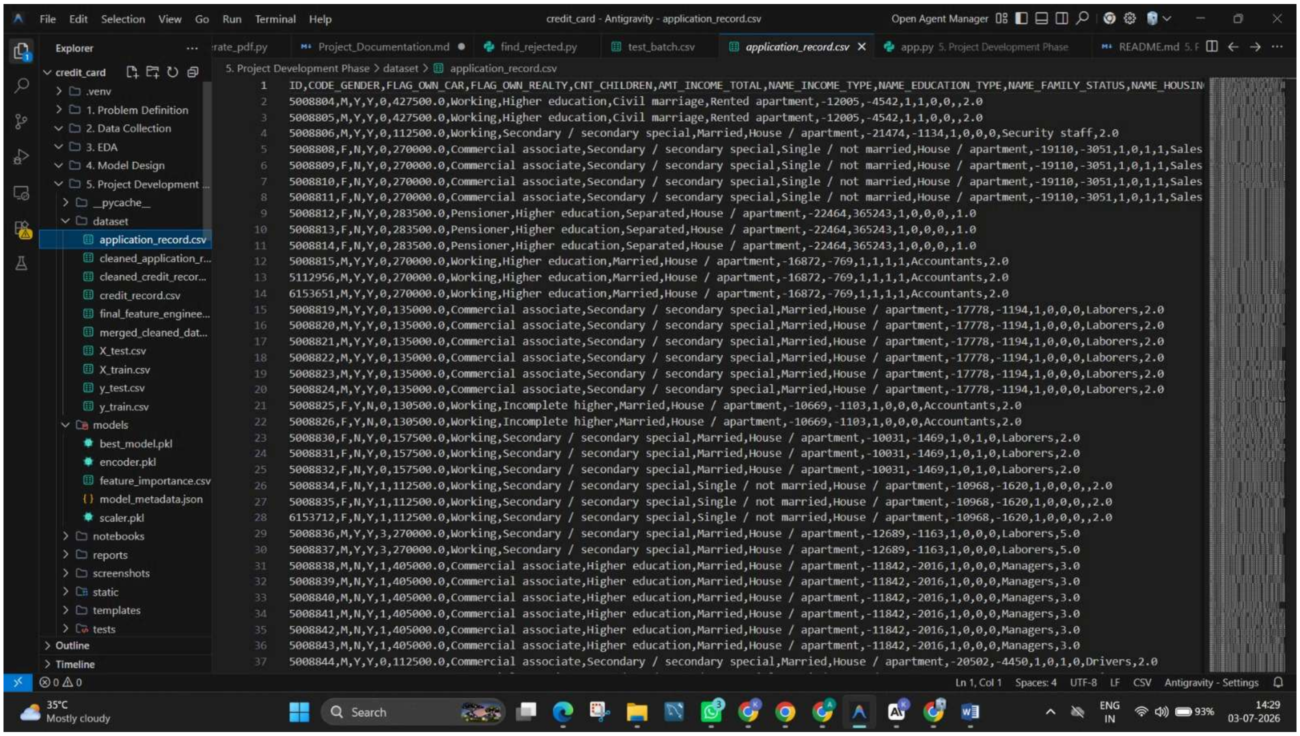
Project Workflow

Project Workflow

Milestone 1: Model Selection and Architecture

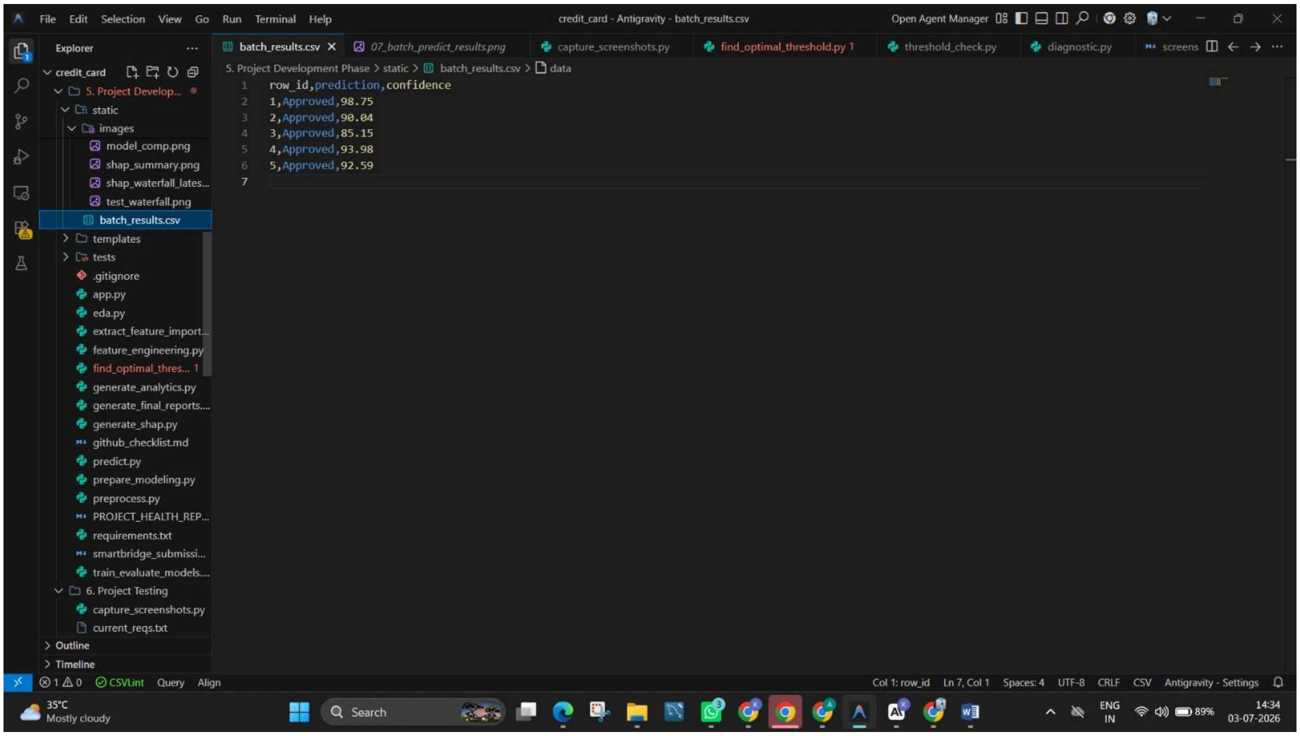
Activity 1.1: Dataset Collection

Collect the Credit Card Approval Prediction dataset containing applicant demographic information, financial details, employment information, and credit history from a reliable source such as Kaggle. The dataset includes applicant records and credit history data that are used for model training and prediction.



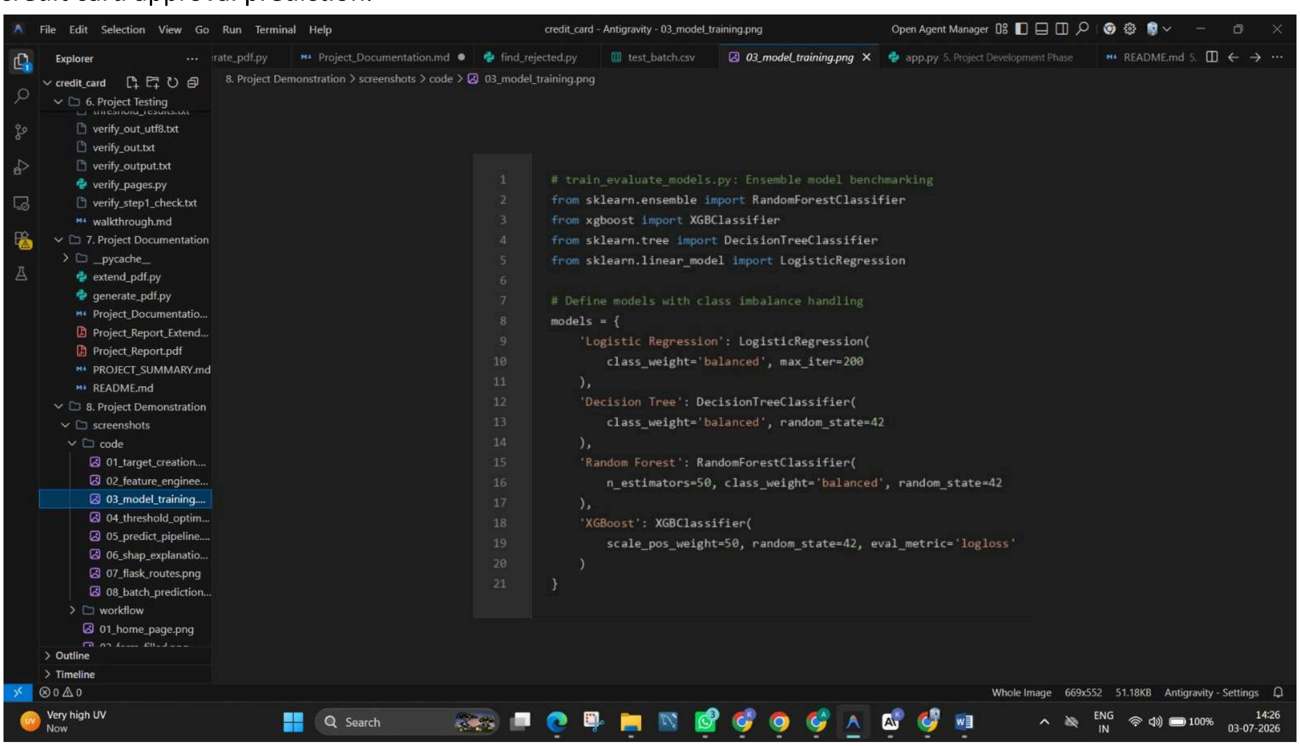
Activity 1.2: Data Preprocessing

Handle missing values, remove duplicate records, merge applicant and credit history datasets, encode categorical variables, perform feature engineering, normalize numerical features, and split the dataset into training and testing sets for machine learning model development.



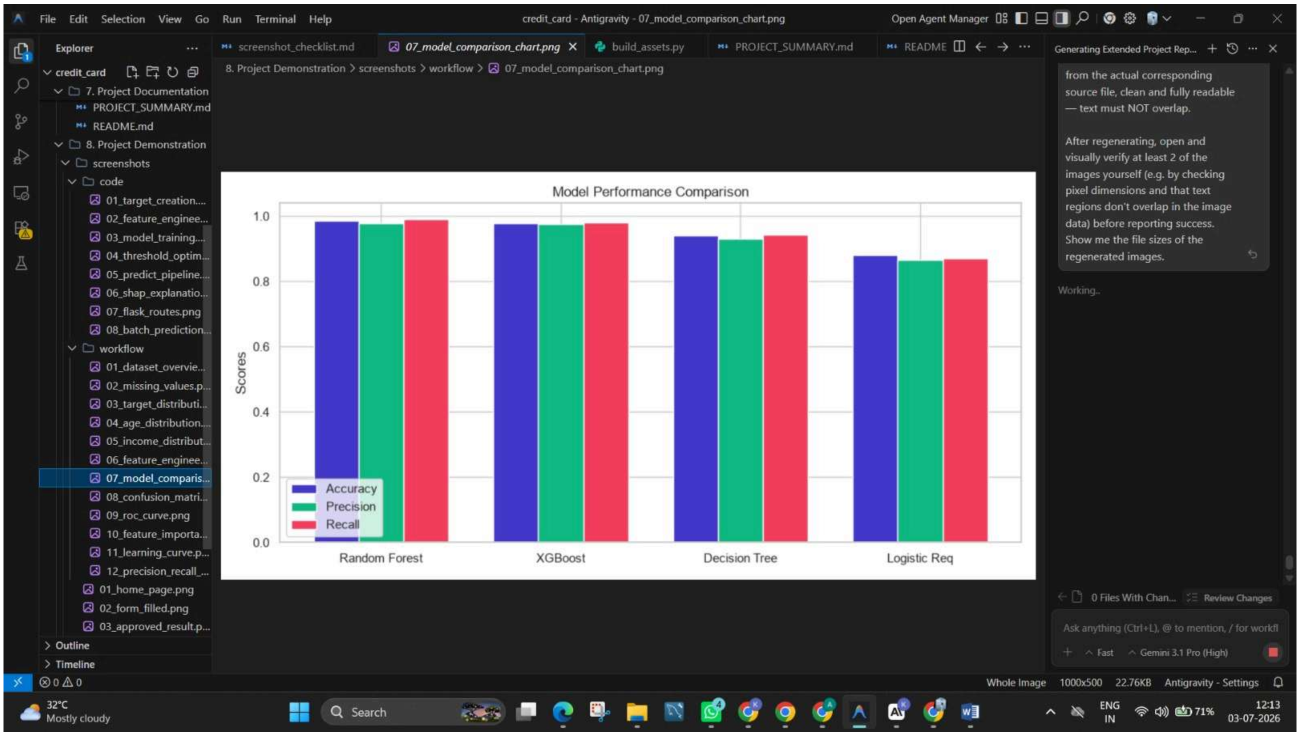
Activity 1.3: Model Training

Train multiple machine learning models such as Logistic Regression, Decision Tree, Random Forest, and XGBoost. Compare their performance using evaluation metrics and select the best-performing model for credit card approval prediction.



Activity 1.4: Model Selection

Compare model performance using Accuracy Score and select the best-performing model from Logistic Regression, Decision Tree, Random Forest, and XGBoost for credit card approval prediction.



Milestone 2: Core Functionalities Development Activity 2.1

Develop User Registration and Login Module.

Activity 2.2

Implement Credit Card Approval Prediction Module.

Activity 2.3

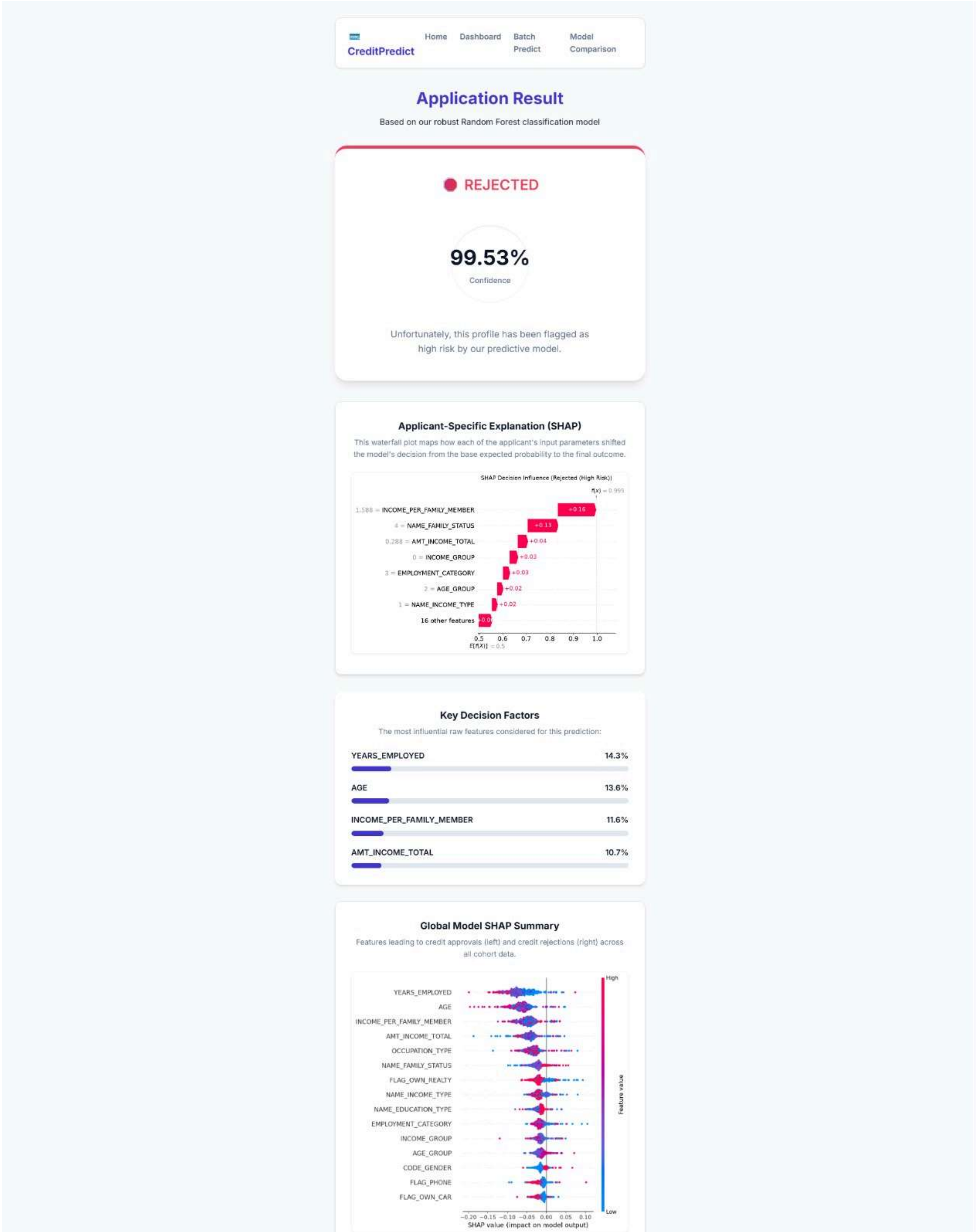
Validate applicant information before prediction.

Activity 2.4

Store prediction results in SQLite Database.

Activity 2.5

Display the credit card approval prediction result.

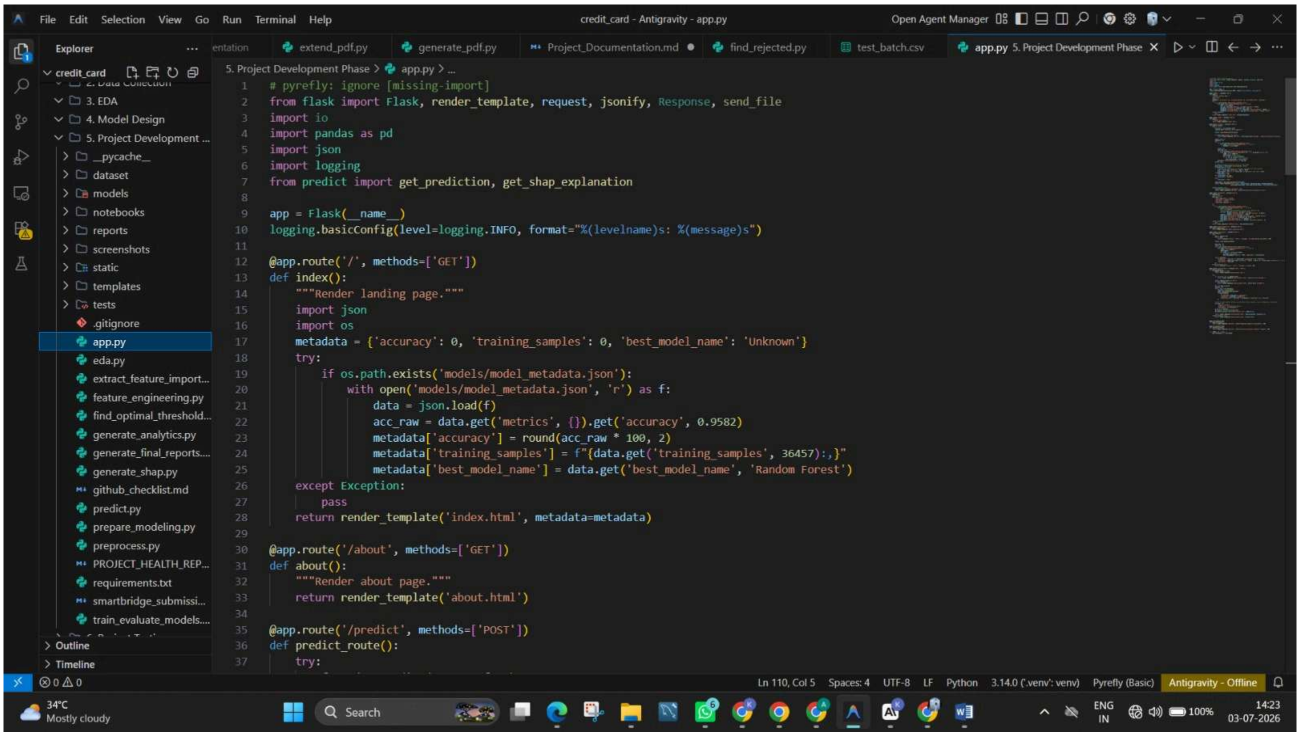


Milestone 3: Flask Application Development Activity 3.1

Create Flask routes for

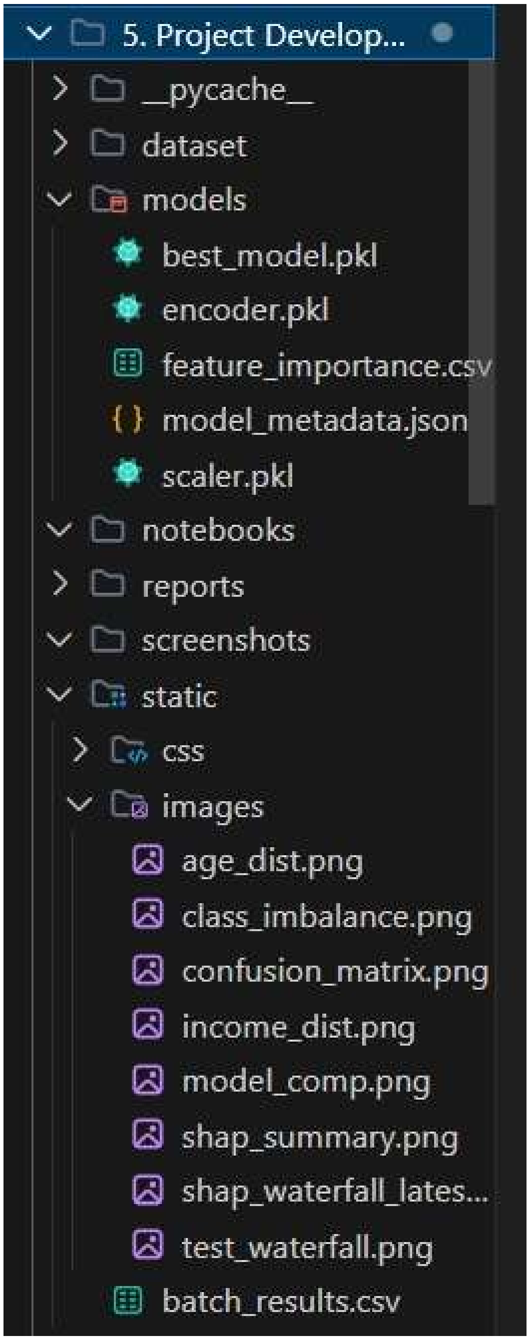
- Home Page

- Login
- Register
- Credit Card Approval Prediction
- Dashboard
- Logout



Activity 3.2

Load the trained best-performing machine learning model using Pickle.

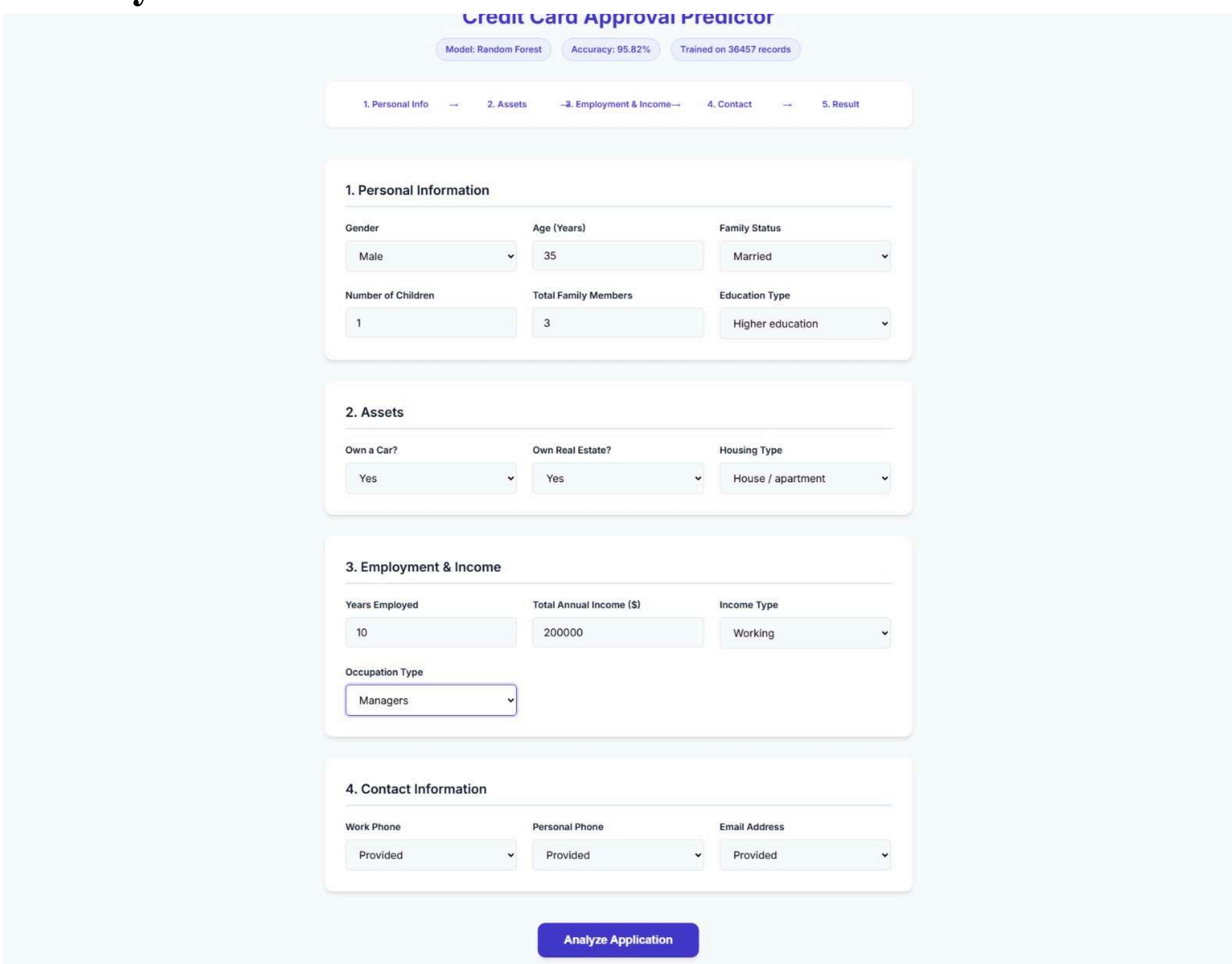


Activity 3.3
Accept applicant information from HTML form.

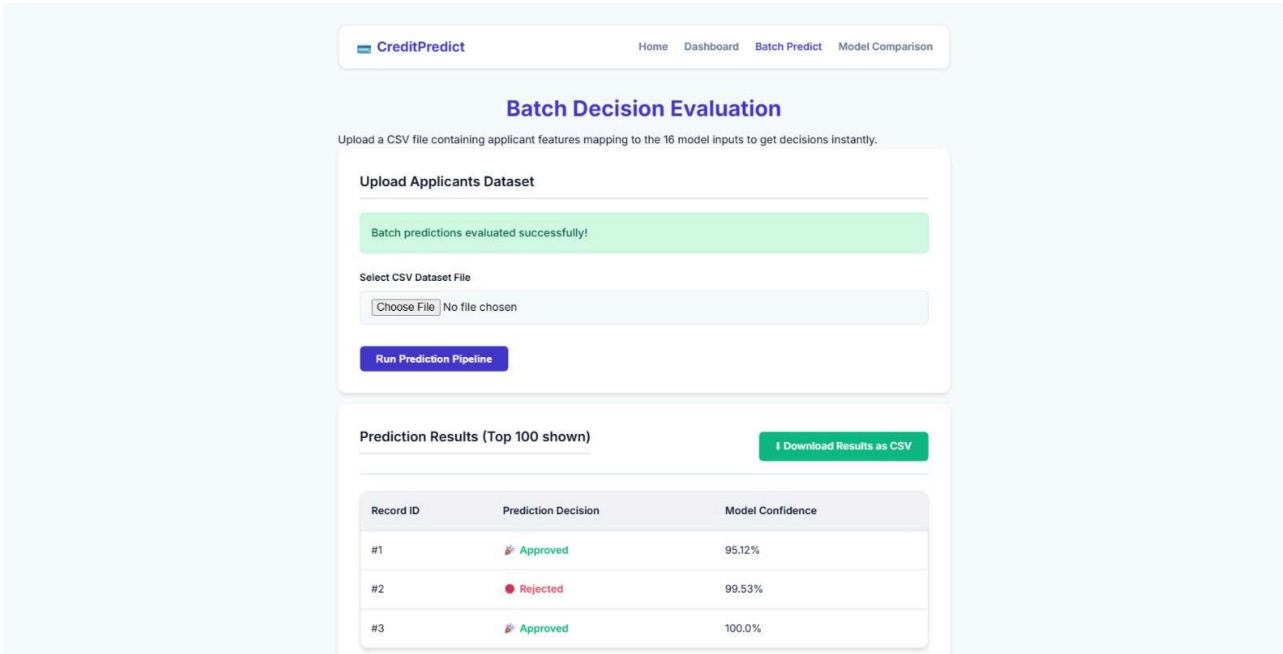
Activity 3.4
Predict credit card approval using the trained model.
Activity 3.5
Store prediction details inside SQLite Database.

	id	applicant...	gender	married	depende...	education	self_emp...	applicant...	coapplic...
1	1	pavan	1.0	1.0	1.0	1.0	1.0	2000	30000
2	2	king	0.0	0.0	2.0	1.0	1.0	10000	20000
3	3	ravi	1.0	1.0	0.0	1.0	1.0	10000	20000
4	4	ravi	1.0	1.0	0.0	1.0	1.0	10000	20000
5	5	siddu	1.0	1.0	1.0	1.0	1.0	10000	200000
6	6	pavan	1.0	1.0	0.0	1.0	1.0	122000	10000
7	7	king	0.0	0.0	1.0	0.0	1.0	2000	10000
8	8	king	0.0	0.0	1.0	0.0	1.0	2000	10000
9	9	siddu	1.0	1.0	0.0	1.0	1.0	10000	2000
10	10	sagar	1.0	1.0	2.0	1.0	1.0	50000	100000

Milestone 4: Frontend Development
Activity 4.1



Activity 4.2



Activity 4.3

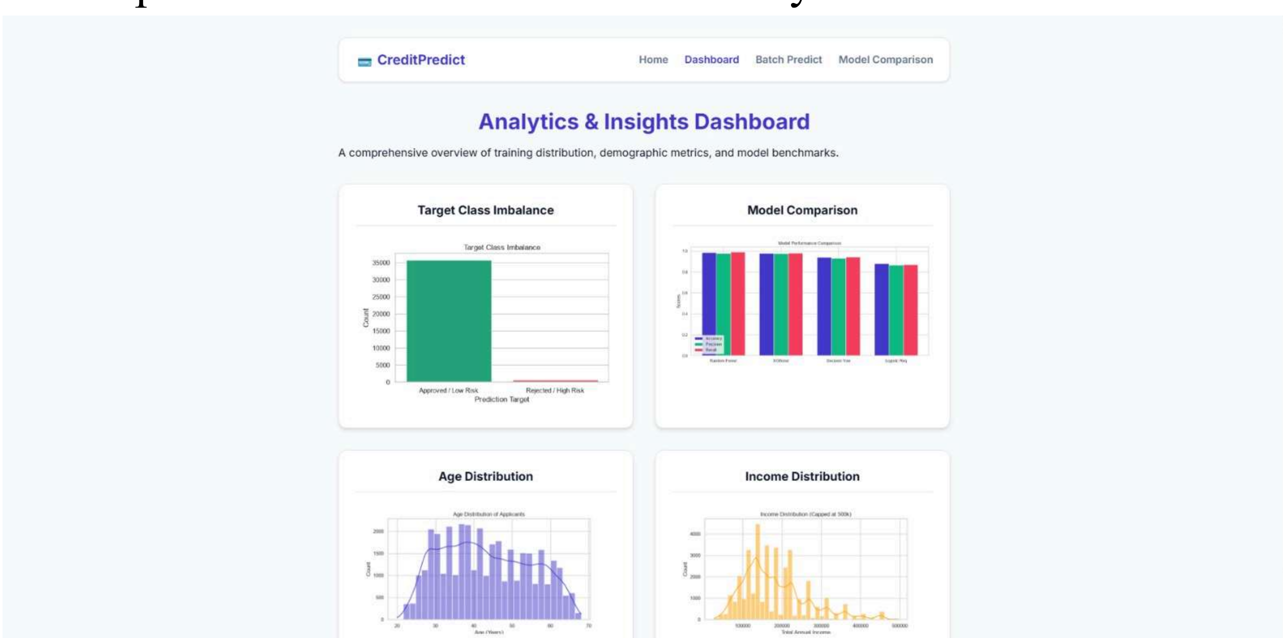
Develop Credit Card Application Form.

Activity 4.4

Display Credit Card Approval Prediction Results.

Activity 4.5

Develop Dashboard for Prediction History.



Activity 5.1

Create Virtual Environment

```
python -m venv venv
```

Activate Environment

```
venv\Scripts\activate
```

Install Dependencies

```
pip install -r requirements.txt
```

Activity 5.2

Run Flask Application

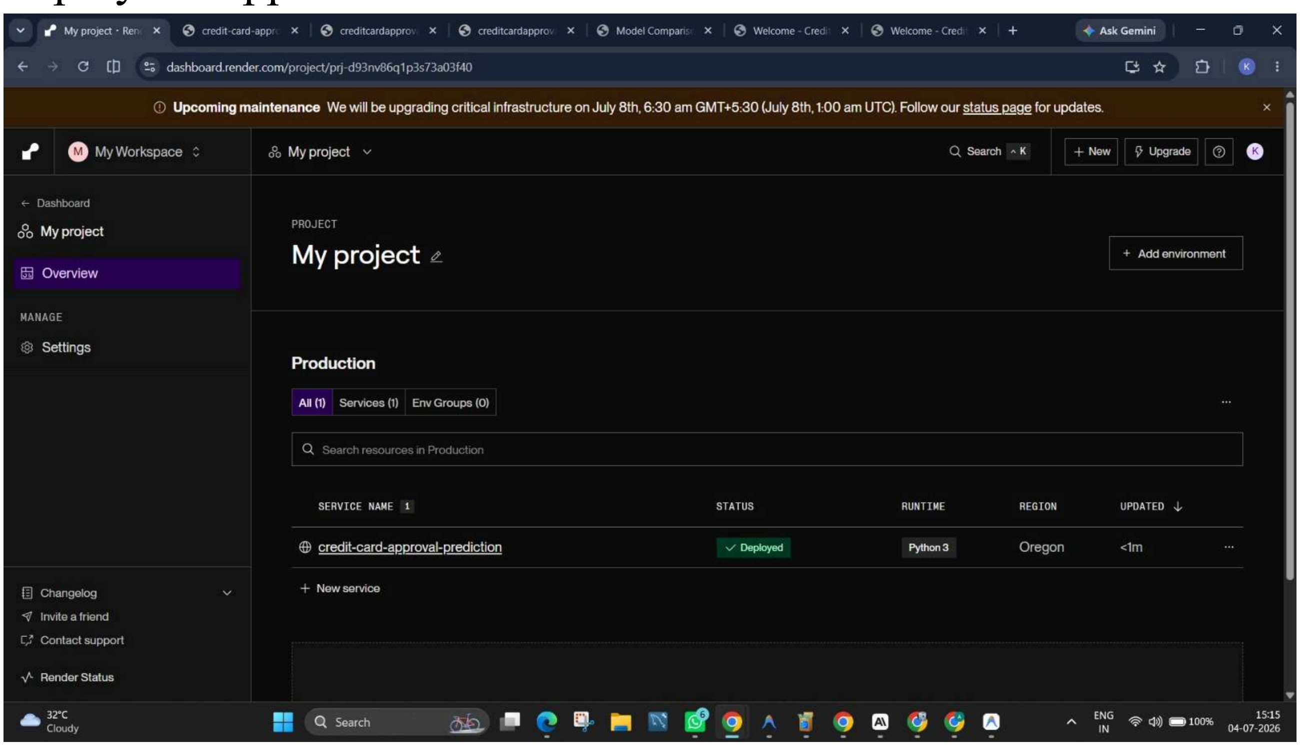
```
python app.py
```

Activity 5.3

Push Source Code to GitHub.

Activity 5.4

Deploy the application on Render.



https://creditcardapprovalprediction.onrender.com

Conclusion:

Credit Card Approval Prediction is an AI-powered web application that helps banks and financial institutions make faster and more accurate credit card approval decisions. By using machine learning algorithms such as Logistic Regression, Decision Tree, Random Forest, and XGBoost, the system predicts whether an applicant is eligible for a credit card.

The project integrates Machine Learning, Flask, HTML, CSS, Bootstrap, JavaScript, SQLite, and IBM Watson Machine Learning into a user-friendly web application. It reduces manual effort, improves decision-making accuracy, and speeds up the credit card approval process.
Future enhancements include cloud database integration, fraud detection, real-time analytics, and improved machine learning models for higher prediction accuracy.